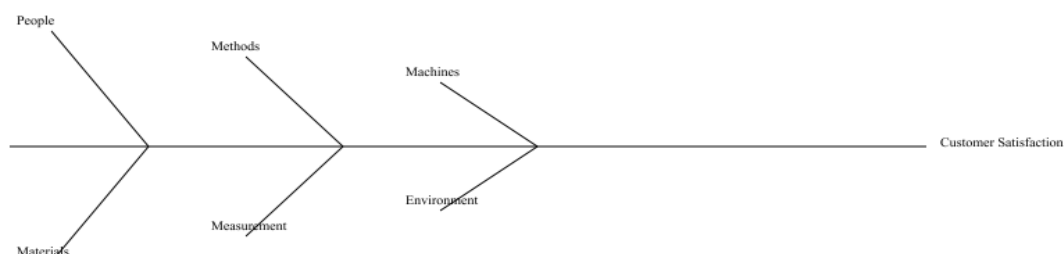


Fishbone Diagram Analysis in Manufacturing: Human-Centered Improvement and the Limits of Artificial Intelligence



Abstract

[illegible]

Introduction

Manufacturing performance is often discussed through the lenses of automation, AI, and digital transformation. However, classical quality management theory argues that sustainable excellence originates from disciplined processes guided by engaged employees. Manufacturing performance is often discussed through the lenses of automation, AI, and digital transformation. However, classical quality management theory argues that sustainable excellence originates from disciplined

[illegible][illegible]

Deming, Juran, Ishikawa, and Ohno consistently emphasized systems thinking, employee participation, process control, and continuous improvement. Their combined work provides the intellectual foundation for root-cause analysis. Deming, Juran, Ishikawa, and Ohno consistently emphasized systems thinking, employee participation, process control, and continuous improvement. Their combined work provides the intellectual foundation for root-cause analysis.

Fishbone Framework

[illegible]

Critical Analysis of AI

[illegible]

detection, and classification. Yet optimization is not equivalent to innovation. AI generally improves an existing process without questioning whether the process itself creates value. AI excels at pattern recognition, optimization, forecasting, anomaly detection, and classification. Yet optimization is not equivalent to innovation. AI generally improves an existing process without questioning whether the process itself creates value. AI excels at pattern recognition, optimization, forecasting, anomaly detection, and classification. Yet optimization is not equivalent to innovation. AI generally improves an existing process without questioning whether the process itself creates value.

The Critical Thinking Gap

[illegible]

or poor communication.

Tacit Knowledge

[illegible]

Implications for Industry 4.0

[illegible]

Conclusion

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